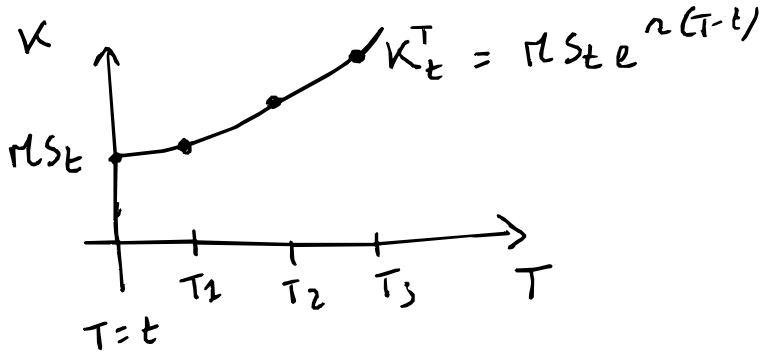
lecture – Thursday 20/02/2025

- <u>slide 3</u>: if moneyness M is fixed,

then

$$K = K_t^T = M \cdot F_t^T = \underbrace{M \cdot S_t e^{r(T-t)}}_{K_t^T}$$



Different strikes = different options

▷ slide 4 : no arb condition on Call prices

proof of condition i): convexity in K

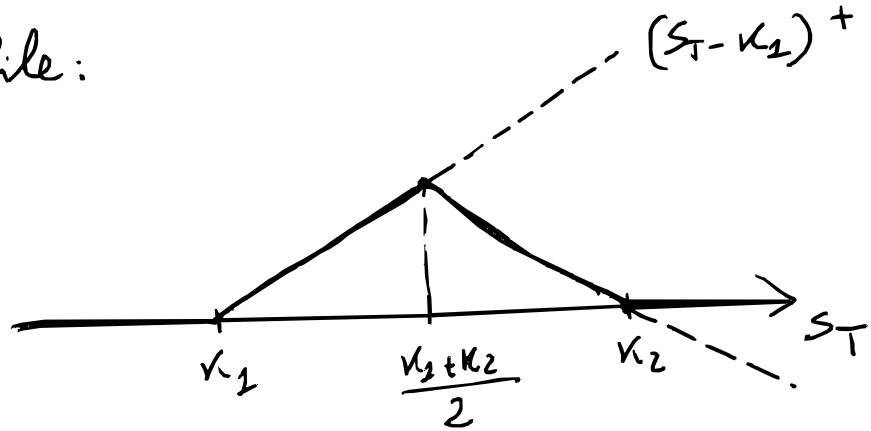
- We consider a combination of 3 call options with same maturity T and different strikes

$$K_1, \quad K_2, \quad \frac{K_1+K_2}{2} \qquad \text{with } K_2 > K_1$$

- The combination of the payoffs is:

$$B(S_T) := (S_T - K_1)^+ - 2\left(S_T - \frac{K_1+K_2}{2}\right)^+ + (S_T - K_2)^+$$

- Payoff profile:



$B(S_T)$: a "Butterfly" or "Butterfly spread"

We note that we have: $\boxed{B(S_T) \geq 0}$ always for every value of S_T

- No arbitrage:

$$B(S_T) \geq 0 \quad \forall S_T \implies \text{price}_t(B(S_T)) \geq 0 \qquad \forall t \leq T$$

- Then, by linearity of prices:

$$\text{price}_t\left((S_T - K_1)^+ - 2\left(S_T - \frac{K_1+K_2}{2}\right)^+ + (S_T - K_2)^+ \right)$$

$$= \text{Call}_t(T, K_1) - 2\,\text{Call}_t\left(T, \frac{K_1+K_2}{2}\right) + \text{Call}_t(T, K_2)$$

$$\geq 0$$

↳ this is precisely the convexity inequality for the fct $K \mapsto \text{Call}_t(T,K)$ at the points K_1, K_2

$$\text{Call}_t\left(T, \frac{K_1+K_2}{2}\right) \leq \frac{1}{2}\text{Call}(T, K_1) + \frac{1}{2}\text{Call}(T, K_2)$$

("the prices of Butterflies are positive")

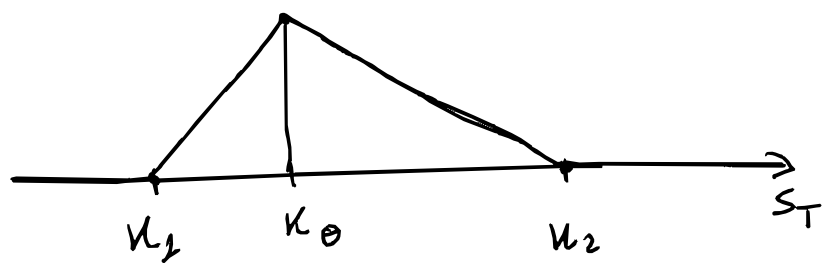
<u>Exercise:</u> with the same argument, show that

$$\text{Call}_t(T, \theta K_1 + (1-\theta)K_2)$$

$$\leq \theta\,\text{Call}_t(T, K_1) + (1-\theta)\,\text{Call}_t(T, K_2)$$

$$\forall \theta \in [0,1], \quad \forall K_1 < K_2$$

The payoff in this case is:



Conclusion: the fct $K \mapsto \mathrm{Call}_t(T, K)$ is convex, $\forall T$

iii) <u>Monotonicity in T</u>: another arbitrage argument, taking into account options with different

We need to use the combination:

$$\left(S_{T_2} - M F_t^{T_2}\right)^+ - \left(S_{T_1} - M F_t^{T_1}\right)^+$$: this is called "calendar spread", $T_2 > T_1$

$$\rightarrow \mathrm{Call}_t(T_2, M F^{T_2}) - \mathrm{Call}_t(T_1, M F^{T_1}) \geq 0$$

$\forall T_2 > T_1$, $\forall$ fixed M $\square$

We have the following

<u>representation result</u> for option prices:

▷ For fixed T, a given function $K \longmapsto C(K)$

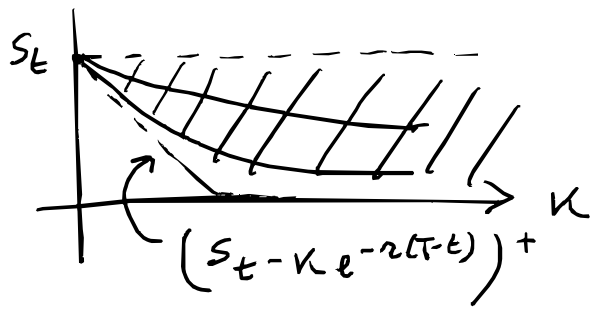
satisfies the conditions

0) $\longrightarrow$ bounds

i) $\longrightarrow$ convexity in K

and the additional property

iii) $\lim_{K \to \infty} C(K) = 0$



<u>if and only if</u> there exists a pricing measure, that is

a probability measure $\mathbb{P}_T$ on $\mathbb{R}_+$ such that

$$C(K) = e^{-r(T-t)} \int_{\mathbb{R}_+} (x-K)^+ \, \mathbb{P}_T(dx)$$

$$= e^{-r(T-t)} \, E\left[(X-K)^+ \right] \quad , \forall K \quad (*)$$

where X is a random variable with distribution $\mathbb{P}_T$.

▷ For Black-Scholes option prices $C(K) = \text{Call}_{BS}(T,K)$, we have seen that

$$X = S_t \, e^{\sigma\sqrt{T-t}\, G - \frac{1}{2}\sigma^2(T-t) + r(T-t)}$$

where $G \sim N(0,1)$

↳ in this case, X has log-normal distribution.

▷ The representation $(*)$ for call prices is always true, possibly with other random variables X.

▷ <u>No arbitrage conditions on the implied volatility</u>

Reminder:

$$\text{Call}_{BS}(t, T, S, K, r, \sigma)$$

$$= S\, N\left(\frac{\ln \frac{S e^{r(T-t)}}{K}}{\sigma \sqrt{T-t}} + \frac{\sigma\sqrt{T-t}}{2} \right) - K e^{-r(T-t)} N\left(\frac{\ln \frac{S e^{r(T-t)}}{K}}{\sigma\sqrt{T-t}} - \frac{\sigma\sqrt{T-t}}{2} \right)$$

<u>Rmk</u>: The <u>normalised</u> call price

(normalised = divided by S)

$$\frac{1}{S}\, \text{Call}_{BS}(t, T, S, K, r, \sigma)$$

only depends on two variables:

$$M = \frac{K}{F} \longrightarrow \text{moneyness}$$

$$v = \sigma\sqrt{T-t} \longrightarrow \text{total volatility}$$

→ We can reparametrise Call^{BS} in terms of v

and of $x = \log(M) = \log\left(\frac{K}{F}\right)$ = log - moneyness.

- This yields the normalised BS call price function:

$$c_{BS}(x, v) = N(d_1(x,v)) - e^x N(d_2(x,v))$$

$$\forall x \in \mathbb{R}, \forall v > 0$$

where
$$d_1(x,v) = \frac{-x}{v} + \frac{v}{2}$$
$$d_2(x,v) = -\frac{x}{v} - \frac{v}{2}$$

- The mapping between the two functions is of course:

$$\text{Call}_{BS}(t, T, S, K, r, \sigma) = c_{BS}\left(\ln \frac{K}{Se^{r(T-t)}}, \sigma\sqrt{T-t}\right)$$

▷ Reminder: implied vol $\sigma_t(T,K)$ from the equation

$$\mathrm{Call}_{BS}\left(t, T, S, K, r, \boxed{\sigma_t(T,K)}\right) = \mathrm{Call}_t(T,K)$$

- Equivalently, we can define

$$\boxed{\hat{\sigma}_t(T,x)}$$

in terms of the log-moneyness x, by setting

$$\hat{\sigma}_t(T,x) = \sigma_t(T,K)\Big|_{K = F_t^T e^x}$$

$$\left(\text{since } x = \log\frac{K}{F} \iff K = F e^x\right)$$

▷ One also defines:

- the <u>total implied vol</u>

$$v_t(T,x) = \sqrt{T-t}\,\hat{\sigma}_t(T,x)$$

- the <u>total implied variance</u>

$$w_t(T,x) = v_t(T,x)^2 = (T-t)\,\hat{\sigma}_t^2(T,x)$$

Definition (arbitrage-free implied vol)

We say that a given $\sigma_t(T,K)$ is free of arbitrage for all $T \geq t$ and $K > 0$ if the function

$$(T,K) \longmapsto \mathrm{Call}_{BS}(t,T,S,K,\ \sigma_t(T,K))$$

satisfies the conditions

0) $\downarrow$ bounds

i) $\downarrow$ convexity in K

ii) $\downarrow$ monotonicity in T

and

iii) $\downarrow$ $\lim_{K\to\infty} = 0$

▷ Equivalently, in terms of $\hat{\sigma}_t(T,x)$:

the function

$$(T,K) \longmapsto \mathrm{Call}_{BS}\left(t,T,S,K,\ \hat{\sigma}_t\left(T,\ \ln\frac{K}{F}\right)\right)$$

satisfies the conditions 0), i), ii), and iii)

<u>Question:</u> more explicit conditions directly expressed in terms of the function $(T,K) \mapsto \sigma_t(T,K)$?

<u>Theorem</u> (no arbitrage conditions on an implied vol surface)

Let $t \geq 0$ be fixed

Let $\hat{\sigma}_t(T,x)$ be a function such that

- $\hat{\sigma}_t(T,x) > 0 \quad \forall T, \forall x$
- $x \mapsto \hat{\sigma}_t(T,x) \in C^2(\mathbb{R}) \quad , \forall T.$

Then, $\hat{\sigma}_t$ is an implied vol free of arbitrage <u>if and only if</u> the corresponding total vol function satisfies the conditions:

$$v_t(T,x) = \sqrt{T-t}\ \hat{\sigma}_t(T,x)$$

1) $$\boxed{\partial_{xx} v_t(T,x) + v_t(T,x)\ \frac{d}{dx} d_1(x, v_t(T,x))\ \frac{d}{dx} d_2(x, v(T,x)) \geq 0}$$

$\forall x \in \mathbb{R}, \forall T$

In a more compact manner:

$$\boxed{v'' + v\ d_1'\ d_2' \geq 0}$$

where we set

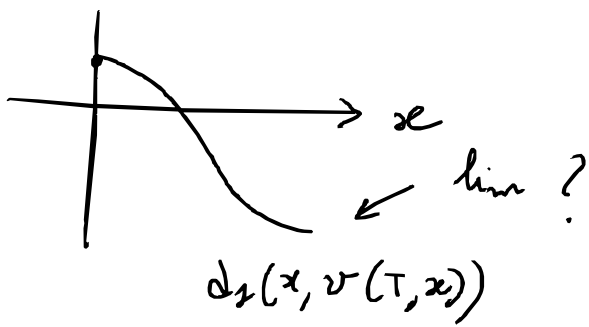
$$\left.\begin{aligned} v''(x) &:= \partial_{xx} v(T,x) \\ d_{1,2}(x) &:= -\frac{x}{v(T,x)} \pm \frac{v(T,x)}{2} \end{aligned}\right\} T \text{ fixed}$$

2) The total vol is increasing in T:

$$T \longmapsto v_t(T, x) \text{ is increasing}, \ \forall x \in \mathbb{R}$$

3) $\lim_{x \to \infty} d_1(x, v_t(T, x))$

$$= \lim_{x \to \infty} \left(- \frac{x}{v_t(T,x)} + \frac{v_t(T,x)}{2} \right) = -\infty$$



<u>Proof</u>: done by showing that

(1) $\Leftrightarrow$ (i) convexity in K of call price

(2) $\Leftrightarrow$ (ii) monotonicity in T ——— with fixed moneyness

(3) $\Leftrightarrow$ (iii) $\lim_{K \to \infty} \text{Call}(T, K) = 0$.

<u>Final comment</u>:

- If we develop the derivatives d_1' and d_2', the condition

1) $\rightarrow$ $v'' + v\, d_1' d_2'$ becomes

$$\left(\mathcal{L} v_t(T, \cdot)\right)(x) \geq 0 \qquad \forall x \in \mathbb{R}$$

where

$$(\mathcal{L} v)(x) = v'' + \frac{1}{v} - 2x \frac{v'}{v} + \frac{(v')^2}{4}\left(\frac{x^2}{v^2} - \frac{v^2}{4}\right)$$

$\hookrightarrow$ 2n order non-linear differential operator applied to the total volatility v.